



Applied Artificial Intelligence

An International Journal

ISSN: 0883-9514 (Print) 1087-6545 (Online) Journal homepage: www.tandfonline.com/journals/uaai20

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To cite this article: Ntombikayise Betreace Tuge & Nkosikhona Theoren Msweli (2026) Explainable Artificial Intelligence in the Banking Sector: A Systematic Literature Review, Applied Artificial Intelligence, 40:1, 2676353, DOI: [10.1080/08839514.2026.2676353](https://doi.org/10.1080/08839514.2026.2676353)

To link to this article: <https://doi.org/10.1080/08839514.2026.2676353>



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Explainable Artificial Intelligence in the Banking Sector: A Systematic Literature Review

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ABSTRACT

This systematic literature review investigates the critical implementation gap between the technical promise of Explainable Artificial Intelligence (XAI) and its operational deployment within the banking sector. While the existing literature extensively documents XAI's potential to enhance transparency, regulatory compliance, and trust, significant barriers prevent its effective integration into banking workflows. Adhering to the PRISMA 2020 methodology, this review synthesizes 78 peer-reviewed studies and industry reports (2021–2025) to analyze why XAI adoption remains challenging despite recognized necessity. Our findings reveal that technical challenges, particularly the accuracy-interpretability trade-off and the “explaining the explanation” problem, are compounded by more fundamental organizational and cultural barriers. These include resistance to change, misalignment between technical teams and business units, and a lack of standardized evaluation frameworks. Crucially, we identify that successful XAI implementation requires addressing socio-technical tensions through interdisciplinary collaboration, context-aware explanation design, and human-centric governance frameworks. This review offers a critical synthesis that moves beyond cataloging XAI techniques to diagnose implementation failures and propose a pathway to bridge the gap between XAI research and banking practice. We conclude that XAI must evolve from a technical add-on to an embedded organizational capability, necessitating structural changes in how banks approach AI governance, stakeholder communication, and ethical oversight.

Introduction

The banking sector's rapid adoption of Artificial Intelligence (AI) has transformed operations across customer service and risk management (Ben et al. 2021; Rajesh 2021). AI technologies now power critical functions, including fraud detection, credit scoring, and personalized financial services, offering unprecedented efficiency and customer insights (Nallakuruppan 2024). This transformation aligns with the broader FinTech evolution, in which algorithmic systems increasingly mediate financial decisions and relationships (Kou, Lu, and Peng 2025).

However, the sophisticated machine learning models driving this transformation – particularly deep neural networks and ensemble methods – operate as “black boxes” with opaque decision-making processes (Pérez-Cruz et al. 2025; van der Burgt 2020). This opacity creates significant challenges for regulatory compliance, ethical accountability, and stakeholder trust. When AI systems deny loans or flag transactions without explanation, they undermine the foundational principles of transparency and fairness in financial services (Khan 2024).

Explainable Artificial Intelligence (XAI) has emerged as a promising solution, comprising techniques that make AI decisions interpretable to human stakeholders (Adadi and Berrada 2018). The field has matured considerably since its emergence, with numerous technical approaches, including model-agnostic methods (e.g., LIME, SHAP), interpretable models, and visualization techniques (Barredo Arrieta et al. 2020; Guidotti et al. 2018).

Despite this technical progress and regulatory pressure from frameworks such as GDPR's “right to explanation,” a persistent implementation gap remains in banking. Previous literature reviews have either

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 Supplemental data for this article can be accessed online at <https://doi.org/10.1080/08839514.2026.2676353>

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focused on technical XAI methodologies broadly (Adadi and Berrada 2018) or AI applications in finance generally (Kudelić, Šmaguc, and Robinson 2025), but none have systematically examined why XAI adoption in banking remains challenging despite clear technical solutions and regulatory imperatives. This review addresses the gap by investigating the socio-technical barriers to XAI integration and identifying pathways for effective implementation.

The primary research question guiding this review is: What are the key barriers to effective XAI implementation in banking, and how can they be addressed through technical, organizational, and regulatory approaches?

Supporting sub-questions include:

- (1) What XAI techniques are most applicable to banking contexts, and what are their limitations?
- (2) How do organizational factors influence XAI adoption and effectiveness?
- (3) What frameworks exist for integrating XAI into banking governance structures?
- (4) How can XAI explanations be tailored to different stakeholder needs in banking?

This systematic review contributes to both academic research and industry practice by providing a comprehensive analysis of XAI implementation challenges in banking, synthesizing solutions across technical and organizational domains, and proposing an integrated framework for overcoming adoption barriers.

Methodology

Overview

This systematic literature review follows the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines (Page et al 2021) to ensure methodological rigor, transparency, and reproducibility. The approach enables comprehensive identification, selection, and critical appraisal of relevant literature while minimizing bias.

Search strategy and information sources

A systematic search was conducted across five major academic databases—IEEE Xplore, ScienceDirect, Scopus, ACM Digital Library, and Google Scholar – using Boolean operators to maximize precision and recall. The search string was: (*“explainable AI”* OR *“interpretable AI”* OR XAI OR *“explainable artificial intelligence”*) AND (bank* OR *“financial service”* OR *“credit score”* OR *“fraud detect”* OR *“risk management”* OR *“regulatory compliance”*) AND (*“implementation”* OR *“adoption”* OR *“challenge”* OR *“barrier*”*).

The search was limited to English-language peer-reviewed journal articles, conference papers, and reputable industry reports published between January 2021 and July 2025. This timeframe captures the period of rapid XAI maturation following foundational works from 2018 to 2020, ensuring the review reflects current developments.

Eligibility criteria

Studies were included if they: (1) focused specifically on XAI applications or implementation in banking/financial services; (2) addressed implementation challenges, adoption barriers, or evaluation frameworks; (3) presented empirical findings or conceptual frameworks with clear methodology; (4) were published within the specified timeframe.

Exclusion criteria included: (1) general XAI surveys without banking context; (2) technical papers focusing solely on algorithmic improvements without implementation considerations; (3) opinion pieces without empirical or conceptual foundation; (4) non-English publications.

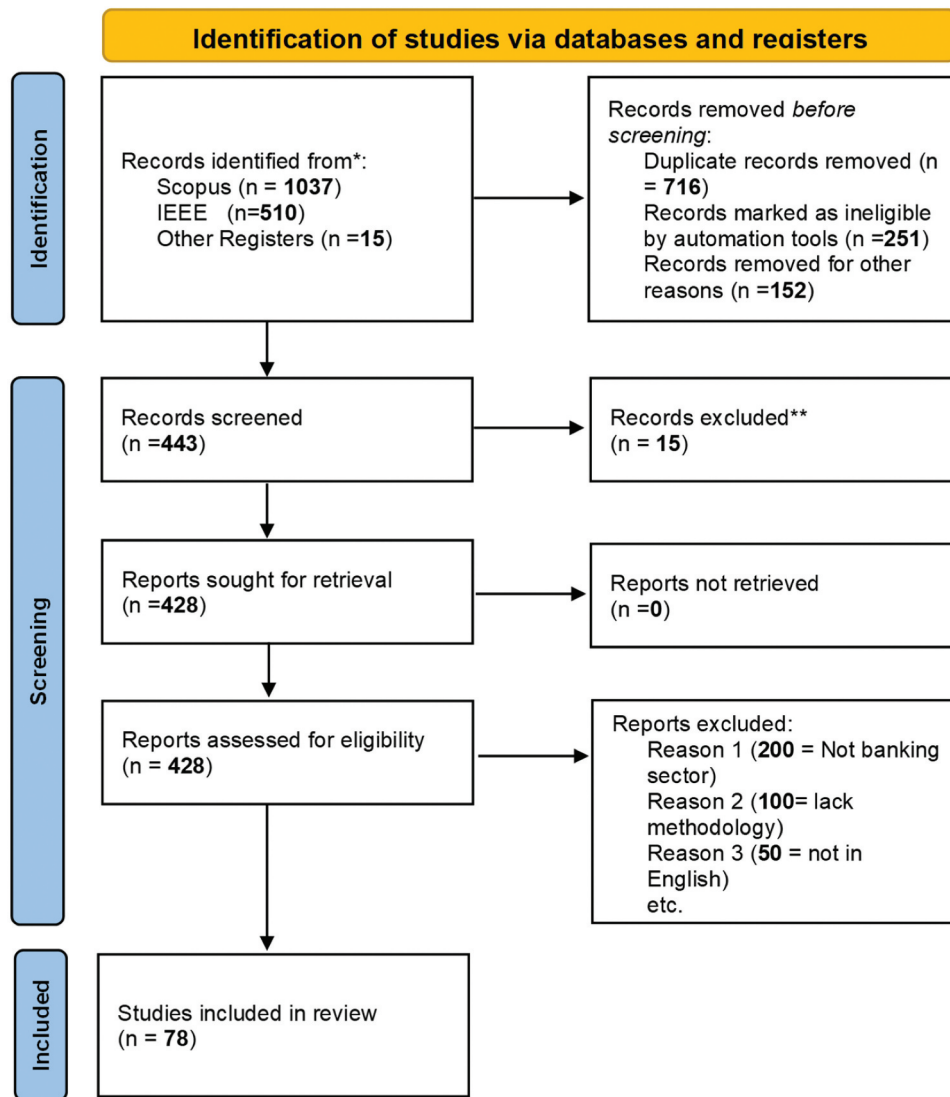


Figure 1. PRISMA flow diagram.

Study selection process

The study selection process followed PRISMA 2020 guidance (Page et al. 2021). Figure 1 presents the PRISMA flow diagram summarizing the identification, screening, eligibility assessment, and inclusion of studies in this review. Initial database searches yielded 1,547 records, with 15 additional records identified through citation tracking. After duplicate removal and title/abstract screening, 428 studies proceeded to full-text review. Two reviewers independently assessed eligibility and resolved discrepancies through consensus discussion. Backward and forward citation tracking of included studies ensured comprehensive coverage. The final corpus included 78 studies for data extraction and synthesis.

Data extraction and quality assessment

Data extraction followed a structured protocol that captured study characteristics (authors, year, methodology); XAI techniques examined; banking applications addressed; implementation challenges identified; proposed solutions; and stakeholder perspectives considered. Quality assessment used adapted Critical Appraisal Skills Program (CASP) checklists to evaluate methodological rigor, relevance to the research questions, and contribution to the field.

Data synthesis and analysis

A hybrid thematic synthesis approach combined deductive coding (based on research questions) with inductive analysis to identify emergent themes. Analysis proceeded in three stages: (1) descriptive synthesis of study characteristics; (2) thematic analysis of challenges and solutions; (3) comparative analysis across stakeholder perspectives and banking domains. NVivo software supported coding and theme development, with regular validation through researcher triangulation.

Results and analysis

Bibliometric overview

Analysis of the 78 included studies reveals accelerating interest in banking XAI, with 65% published in 2023–2025. Geographically, research concentration correlates with regulatory stringency, with 42% of studies from the European Union (particularly addressing GDPR compliance) and 31% from North America. Methodologically, case studies (45%) and conceptual frameworks (30%) dominate, with limited longitudinal or large-scale empirical studies.

To characterize the intellectual structure of the included literature, a keyword co-occurrence analysis was conducted using VOSviewer. [Figure 2](#) presents the resulting network map, showing how frequently keywords co-occur and how themes cluster within the banking XAI literature (van Eck and Waltman 2010).

The map indicates three broad thematic clusters: (1) technical XAI methods (e.g., SHAP, LIME, interpretable models); (2) banking applications (e.g., credit scoring, fraud detection, risk management);

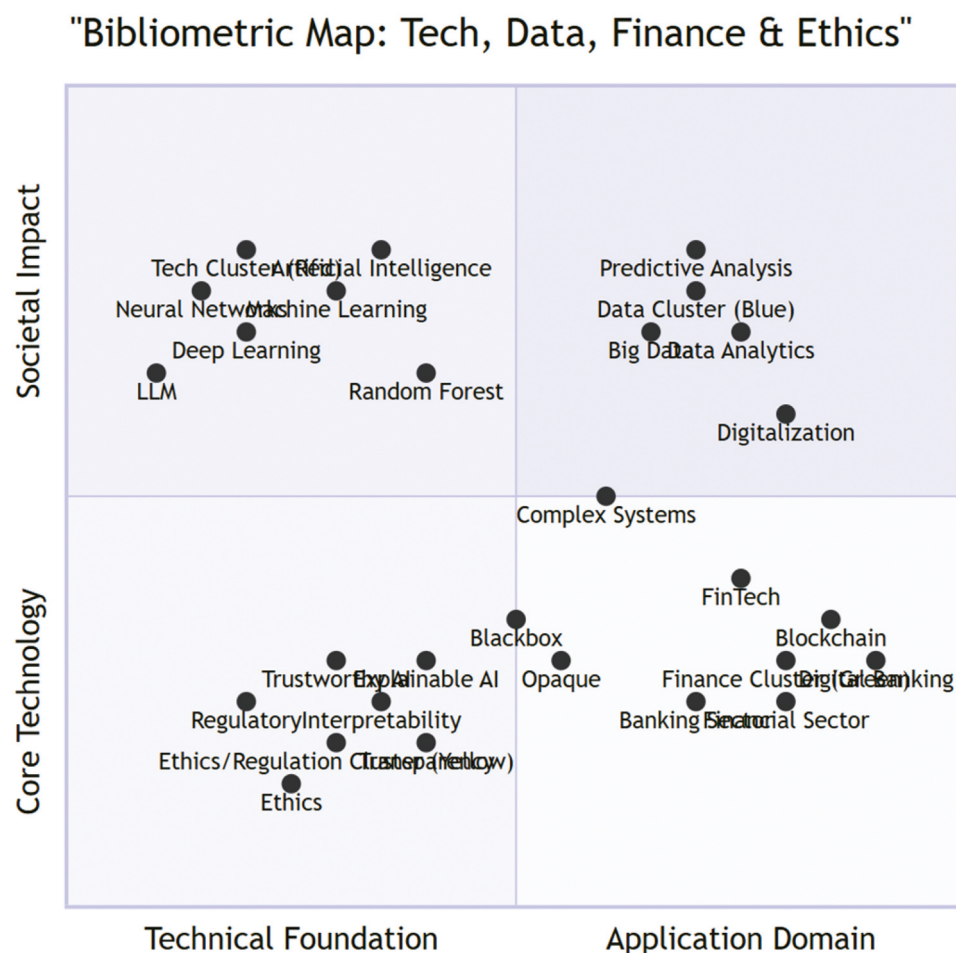


Figure 2. Keyword co-occurrence network of banking XAI literature (VOSviewer). Note. Map generated in VOSviewer; based on van Eck and Waltman (2010).

and (3) governance considerations (e.g., ethics, regulation, accountability). Notably, connections between the technical and governance clusters appear comparatively weak, reinforcing the implementation and integration gap explored in this review.

Cluster analysis

- **Technical Foundation Cluster** (e.g., “Neural Networks,” “Deep Learning,” “LLM”): **Focus:** This cluster embodies the core technological and methodological engines of the field, encompassing advanced algorithms, AI/machine learning techniques such as Random Forest, and foundational concepts like “Complex Systems” and “Core Technology.” **Characteristic:** The inclusion of terms such as “Blackbox” and “Opaque” within or adjacent to this cluster highlights the central challenge driving XAI research: the problem of interpretability in sophisticated models.
- **Finance/Application Domain Cluster** (e.g., “FinTech,” “Banking,” “Credit Scoring”): **Focus:** This cluster covers application and commercialization, including financial sectors, technologies such as Blockchain, and essential banking functions such as risk management and fraud detection. **Characteristic:** It serves as the “customer” for outputs from the technical cluster, deploying predictive analytics and AI models to drive real-world decision-making with significant impact.
- **Ethics/Governance Cluster** (e.g., “Ethics,” “Regulation,” “Trustworthy AI”): **Focus:** This cluster comprises the societal, regulatory, and normative frameworks that address demands for interpretability, accountability, and ethical governance in deploying technology in high-stakes financial domains. **Characteristic:** It responds to the “black box” challenge and the societal impacts that arise from the convergence of technology and finance.
- **Characteristic:** It serves as the “customer” for outputs from the technical cluster, deploying predictive analytics and AI models to drive real-world decision-making with significant impact.
- **Finance/Application Domain Cluster** (e.g., “FinTech,” “Banking,” “Credit Scoring”)

Critical observation: the research gap

The network reveals a limited connection among the technical, ethics, and governance clusters, as illustrated in [Figure 2](#) using VOSviewer analysis by van Eck and Waltman (2010).

Although data-related terms such as “Big Data Analytics” and “Digitalization” bridge the technical and finance clusters, the pathway from core technology to societal impact is not strongly represented. While terms like “Trustworthy AI” and “Regulatory Interpretability” are presented as desired outcomes, they are not robustly linked to specific technical methods such as “SHAP” or “LIME,” underscoring a research gap. As a result, technical XAI research and ethical-governance research often occur in parallel, with limited integration.

The Data/Blue Cluster (“Big Data Analytics,” “Digitalization”) serves as a bridge between technology and finance, demonstrating that data is the common denominator and driving force for both research and application.

However, the connection from “Core Technology” to “Societal Impact” is underdeveloped. The outcomes of “Trustworthy AI” and “Regulatory Interpretability” are present but lack strong ties to specific technical approaches, confirming gaps where technical and governance-focused research remain too separated.

Implications

Several key implications arise for researchers, practitioners, and policymakers:

(1) For Research (Academic Implications)

- **Need for Interdisciplinary Work:** There is a clear need for interdisciplinary projects that co-design technical XAI solutions and governance frameworks from the outset, actively building connections between these clusters.
- **From “Interpretable” to “Actionably Accountable”:** Technical XAI research in finance should be assessed not only for accuracy and fidelity, but also for its ability to fulfill regulatory requirements and ethical standards, such as fairness and the right to explanation.

- **Bridging the Vocabulary Gap:** Researchers must translate technical XAI concepts into the language of law, ethics, and business risk and vice versa to promote effective communication and collaboration.
- (2) For Industry & Regulation (Practical Implications)
- **Strategic Risk for Financial Institutions:** The gap between technical and governance clusters represents a significant operational and compliance risk. Financial institutions deploying opaque AI systems for credit scoring must address regulatory requirements, making investment in integrated explainability solutions an strategic imperative.
 - **Policy and Standard Development:** Regulators and standards organizations need to collaborate with technical experts to develop rules and standards that are both principled and practical. Vague demands for interpretability are insufficient; actionable standards must be established.
 - **New Roles and Skills:** The finance sector must cultivate new roles such as “AI Governance Officer” and “*Ethical AI Auditor*,” along with hybrid skill sets that blend data science, legal, compliance, and ethical expertise.
- (3) For the Broader Field (Field-Shaping Implications)
- **Redefining “Trustworthy AI” in Finance:** Trustworthiness should be understood as a holistic outcome, created at the intersection of robust technology (XAI), validated financial utility, and legitimate governance.
 - **Call for Synthesis:** This review’s focus on bridging the gap between clusters is vital. The field requires further systematic reviews, frameworks, and benchmarks that explicitly encourage and measure integration across technical, financial, and governance domains.

In summary, the bibliometric map exposes a critical disconnect in the research landscape. The responsible and mature deployment of AI in finance relies on actively and collaboratively closing the gap between technologists, practitioners, and those responsible for governance and ethics.

Technical implementation challenges

The literature identifies several persistent technical barriers to the implementation of Explainable Artificial Intelligence (XAI), as summarized in Table 1 below. The table presents commonly used XAI techniques, typical banking applications, key advantages, and implementation challenges. The representative studies included are illustrative rather than exhaustive.

The accuracy-interpretability trade-off remains the most cited technical challenge, particularly for high-stakes applications like credit scoring, where both predictive performance and regulatory compliance are essential (Rudin 2019). This tension often forces banks to choose between powerful “black box” models with post-hoc explanations or inherently interpretable models with potentially lower performance.

The “*explaining the explanation*” problem represents a more subtle but critical challenge: post-hoc explanation methods are themselves complex models whose reliability must be assessed (Doshi-Velez and Kim 2017). In banking contexts, where explanations may be legally contested or scrutinized by regulators, this meta-explanation challenge becomes particularly significant.

Table 1. Technical XAI techniques and their limitations in banking.

XAI Technique	Primary Banking Applications	Key Advantages	Implementation Challenges	Representative Studies
SHAP/SHAP-based methods	Credit scoring, Risk assessment	Mathematical foundation, Consistency guarantees	Computational intensity, Complexity for non-experts	Lundberg and Lee (2017); Ozili (2024)
LIME/Local explanations	Fraud detection, Customer analytics	Model-agnostic, Intuitive visualizations	Instability, Approximation errors	Ribeiro, Singh, and Guestrin (2016); Zong (2022)
Counterfactual explanations	Credit denials, Customer communications	Actionable insights, Intuitive format	Generation complexity, Plurality problem	Wachter, Mittelstadt, and Russell (2017); Rai (2020)
Interpretable models	Regulatory reporting, High-stakes decisions	Built-in transparency, Simplicity	Performance trade-offs, Limited complexity	Rudin (2019); Molnar (2022)
Rule extraction	Compliance documentation, Audit trails	Human-readable format, Verifiability	Fidelity issues, Scalability limitations	Guidotti et al. (2018)

Table 2. Organizational barriers to XAI implementation.

Barrier Category	Specific Challenges	Impact on Implementation	Proposed Solutions
Cultural Resistance	Short-term focus, Risk aversion, Legacy mindset	Delayed adoption, Superficial compliance	Leadership commitment, ROI demonstration, Phased implementation
Resource Constraints	Skills gaps, Budget limitations, Legacy systems	Limited scope, technical debt accumulation	Strategic partnerships, Gradual modernization, Training programs
Measurement Gaps	Lack of XAI evaluation metrics, Unclear success criteria	Difficulty prioritizing initiatives, Unproven value	Development of banking-specific metrics, Case studies

Organizational and Cultural Barriers

Beyond technical challenges, organizational factors pose substantial implementation barriers (see Table 2) (Capuano 2022; Ravikumar et al. 2021).

Resistance to Cultural Change: Multiple studies identify organizational culture as the primary barrier to XAI adoption (Capuano 2022; Ravikumar et al. 2021). Banking institutions often prioritize short-term profitability over long-term investments in transparency infrastructure, particularly when XAI's return on investment (ROI) is difficult to quantify. Risk-averse cultures may view explanatory transparency as increasing rather than mitigating liability.

Interdisciplinary Communication Gaps: The literature consistently identifies communication breakdowns between technical teams (data scientists, AI developers) and business stakeholders (compliance officers, risk managers, executives) (Fernandes, Silva, and Neto 2021; Jensen, Wilson, and Olsson 2023). Technical teams often produce explanations inappropriate for their audience, for example, providing feature importance plots to loan applicants or regulatory auditors without the necessary context or translation.

Skills and Resource Limitations: Effective XAI implementation requires specialized expertise spanning data science, domain knowledge, ethics, and regulatory compliance. Many banks lack this interdisciplinary talent, and training existing staff presents significant challenges (Holmström 2022). Resource constraints are particularly acute for smaller institutions, potentially creating competitive disadvantages.

Table 2 synthesizes organizational and cultural barriers reported in the reviewed studies and summarizes their impacts and commonly proposed mitigation strategies.

Stakeholder-Specific Explanation Needs

A critical finding across studies is the diversity of explanation requirements among different banking stakeholders:

Regulators and Auditors require comprehensive, verifiable, and standardized explanations that enable systematic review to ensure compliance with regulations such as GDPR, fair lending laws, and financial stability requirements. These stakeholders prioritize audit trails, documentation of model limitations, and evidence of bias testing (Muslimin 2024; Ozili 2024).

Customers need concise, actionable, and contextual explanations that help them understand decisions affecting their financial lives. For credit applicants, this means clear reasons for denials with specific guidance for improvement. For fraud detection, it means understandable explanations of flagged transactions without overwhelming technical detail (Kalyani 2023; Shin 2021).

Internal Business Users (loan officers, risk managers, compliance staff) require explanations tailored to their specific decision-making contexts. These explanations must balance technical accuracy with operational relevance, enabling informed human oversight of AI systems (Remolina 2023; Schmidt and Lee 2023).

Technical Teams need explanations that support model development, debugging, and improvement. Their requirements focus on technical metrics, feature importance analyses, and error diagnosis tools (Molnar 2022; Prasanth et al. 2023).

The literature reveals a significant gap between the explanation formats typically produced by XAI techniques and the diverse needs of these stakeholder groups. Most XAI research focuses on technical users, with limited attention to other stakeholder perspectives.

Regulatory and Ethical Considerations

Regulatory frameworks both enable and constrain the implementation of XAI in banking. GDPR's "*right to explanation*" has been a significant driver of XAI adoption in Europe, though its practical implementation remains contested (Marshadul 2023). Sector-specific regulations in banking, including fair lending laws and anti-money laundering requirements, create additional explanatory demands.

Ethically, XAI serves as both a tool for and a subject of ethical scrutiny. While XAI techniques can help identify and mitigate algorithmic bias (Prasanth et al. 2023), they can also provide a false sense of ethical assurance if explanations are misleading or incomplete (Accenture 2024). The literature emphasizes that XAI is necessary but insufficient for ethical AI; it must be embedded within broader ethical governance frameworks that include human oversight, continuous monitoring, and accountability mechanisms.

Emerging Frameworks for XAI Integration

Several frameworks for XAI integration in banking emerge from the literature, though none have achieved widespread adoption:

The Context-Aware Framework emphasizes starting with stakeholder analysis to determine explanation requirements before selecting technical approaches (Shaardha and Alphy 2021). This approach recognizes that different banking applications and stakeholders need different explanation types, formats, and levels of detail.

The Lifecycle Integration Framework embeds XAI throughout the AI development process rather than treating it as a post-hoc addition (Nallakaruppan 2024). This includes explainability requirements in initial design, interpretability metrics during development, and explanation generation as part of deployment.

The Governance-Centered Framework focuses on organizational structures for XAI oversight, including review boards, documentation standards, and audit processes (Capuano 2022; Page et al. 2021). This approach treats XAI as primarily a governance challenge requiring policy, procedures, and accountability mechanisms.

The Human-Centered Framework prioritizes user experience design for explanations, drawing on principles of human-computer interaction to create understandable, actionable explanations (Nguyen and Lee 2021; Smith and Zhang 2022).

Despite these frameworks, implementation remains largely ad hoc, with few banks reporting comprehensive, enterprise-wide XAI strategies. The literature identifies a need for more integrated approaches that combine technical, organizational, and governance perspectives.

Discussion

Introduction

The implementation of Explainable AI (XAI) in the banking sector presents a critical juncture at which technological capabilities must converge with organizational readiness and regulatory frameworks (Accenture 2024; Capuano 2022; Nallakaruppan 2024; Shaardha and Alphy 2021; Smith and Zhang 2022).

The preceding analysis of technical feasibility, organizational dynamics, and governance structures indicates that achieving effective and responsible AI adoption is not a singular challenge but a multifaceted one (Capuano 2022; Nallakaruppan 2024; Nguyen and Lee 2021; Page et al. 2021; Shaardha and Alphy 2021; Smith and Zhang 2022). This discussion synthesizes these findings, arguing that XAI implementation is inherently an a socio-technical endeavor, where success depends on the coherent integration of algorithms, processes, people, and rules within the unique high-stakes context of finance (Bostrom and Heinen 1977; Černevičienė and Kabašinskas 2024).

The Socio-Technical Nature of XAI Implementation

The synthesis reveals that XAI implementation in banking is fundamentally an a socio-technical challenge requiring integrated solutions across technical, organizational, and regulatory domains. Technical solutions alone cannot overcome organizational resistance or regulatory uncertainty, while organizational initiatives

without technical feasibility are equally ineffective. This aligns with broader information systems research emphasizing the interdependence of social and technical systems in technology adoption (Bostrom and Heinen 1977; Černevičienė and Kabašinskas 2024).

The banking context introduces unique socio-technical complexities due to stringent regulation, public trust requirements, and the high-stakes nature of financial decisions. Unlike other sectors where AI explainability may be primarily a technical concern, in banking it becomes a matter of legal compliance, ethical obligation, liability management, and competitive differentiation. Figure 3 presents the socio-technical framework synthesized from the review, illustrating how the technical domain (models and explanation methods), governance and regulatory domain (policies, auditability, accountability), and organizational and cultural domain (skills, processes, adoption) interact to shape XAI implementation outcomes.

The textual content presented in the figure does not originate from a single externally published academic source. Instead, it represents a conceptual framework developed by the authors from their synthesis of study findings. It visually integrates the key themes and constructs explored in the literature:

- **High-Stakes Environment:** Financial Decisions, Legal Compliance & Liability, Competitive Differentiation.
- **Technical Domain:** XAI Algorithms (LIME, SHAP), Model Transparency, Feasibility & Performance.
- **Governance & Regulatory Domain:** Regulatory Guidelines, Audit Trails, Accountability, Governance Frameworks.
- **Organizational & Cultural Domain:** Cultural Adoption, Skill Gaps & Training, Process Integration, Ethical Principles.

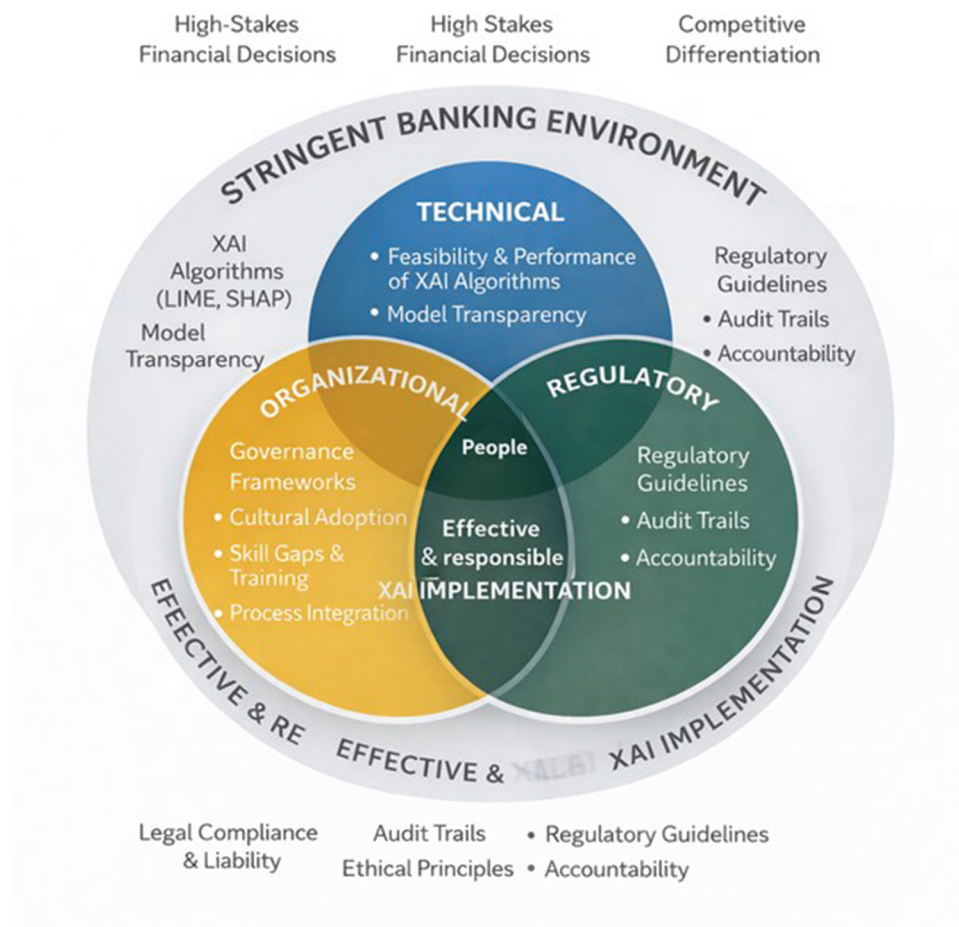


Figure 3. Socio-technical framework for XAI implementation in banking. Note. Figure created by the authors (own synthesis).

The purpose of the framework is to illustrate the interconnectedness of these domains in creating an “EFFECTIVE & RESPONSIBLE” regulatory and operational environment for XAI in banking. It is a figurative representation of the socio-technical framework that underpins the discussion in Introduction.

Bridging the Implementation Gap: An Integrated Framework

Based on our synthesis, we propose an integrated framework for XAI implementation in banking addressing identified gaps:

Phase 1: Contextual Analysis and Stakeholder Mapping

- Identify all stakeholders affected by AI decisions (customers, regulators, internal users)
- Map specific explanation needs for each stakeholder group and use case
- Define success criteria for explanations (comprehensibility, actionability, verifiability)

Phase 2: Technical Design and Selection

- Select XAI techniques appropriate for specific models and stakeholder needs
- Develop evaluation metrics for explanation quality in banking contexts
- Design explanation interfaces tailored to different user groups

Phase 3: Organizational Integration

- Establish cross-functional XAI governance structures
- Develop training programs for different stakeholder groups
- Create documentation and communication protocols for explanations

Phase 4: Continuous Monitoring and Improvement

- Implement feedback mechanisms from explanation consumers
- Regularly audit explanation effectiveness and accuracy
- Update XAI approaches as models, regulations, and stakeholder needs evolve

This framework emphasizes the iterative, context-dependent nature of XAI implementation, rejecting one-size-fits-all approaches in favor of adaptive, stakeholder-centered design.

Addressing the Accuracy-Interpretability Trade-Off in Practice

Our analysis suggests practical approaches to the accuracy-interpretability trade-off:

Contextualized Trade-off Decisions: The appropriate balance between accuracy and interpretability varies by banking application. For high-stakes, heavily regulated decisions like credit approvals, regulatory requirements may justify some accuracy sacrifice for interpretability. For lower-stakes applications like marketing personalization, accuracy may take precedence.

Hybrid Approaches: Combining inherently interpretable models for critical decision components with more complex models for supporting analyses can optimize both accuracy and explainability. For example, using interpretable models for final credit decisions while employing complex models for risk factor identification.

Progressive Explanation: Providing different explanation levels for different stakeholders detailed technical explanations for model validators, summary explanations for loan officers, and simple reasons for customers can address diverse needs without sacrificing model performance.

Measuring XAI Effectiveness and ROI

A significant implementation barrier is the difficulty measuring XAI’s value. Our synthesis suggests several banking-specific metrics:

Compliance Metrics: Reduction in regulatory violations, faster audit completion times, decreased compliance costs.

Operational Metrics: Reduced time for model validation, decreased manual review requirements, fewer customer complaints about AI decisions.

Business Metrics: Improved customer trust and retention, competitive differentiation through transparency, reduced litigation risk.

Ethical Metrics: Demonstrated reduction in algorithmic bias, increased fairness in outcomes across demographic groups.

Developing standardized metrics for XAI effectiveness should be a priority for both researchers and practitioners to enable evidence-based implementation decisions.

Regulatory Evolution and Standardization Needs

Current regulatory approaches to XAI in banking vary significantly across jurisdictions, creating implementation challenges for global institutions. Principles-based regulations like GDPR provide flexibility but also uncertainty about compliance standards. The literature suggests several directions for regulatory evolution:

Industry Standards Development: Collaborative development of XAI standards for banking, like Basel Accords for risk management, could provide clearer implementation guidance while maintaining flexibility.

Regulatory Sandboxes: Controlled environments for testing XAI approaches could help regulators and banks collaboratively develop practical compliance frameworks (Ferreira, Fernandes, and Ferreira 2022; Yeung 2019).

Explanation Certification: Third-party certification of XAI systems, like financial audit practices, could provide independent validation of explanation adequacy.

Future Research Directions

Our synthesis identifies several priority areas for future research:

Longitudinal Studies: Research tracking XAI implementation over time in banking institutions to understand evolution patterns, success factors, and long-term impacts.

Cross-Cultural Comparisons: Comparative studies of XAI adoption across different regulatory environments and banking cultures to identify transferable practices.

Explanation Personalization: Research on adaptive explanation systems that tailor content and format to individual user characteristics, preferences, and contexts.

Integrated Evaluation Frameworks: Development of comprehensive frameworks for evaluating XAI systems across technical, organizational, and ethical dimensions.

Economic Analysis: Cost-benefit studies of XAI implementation in specific banking applications to strengthen business cases for adoption.

Conclusion

This systematic literature review demonstrates that XAI implementation in banking faces multidimensional challenges spanning technical limitations, organizational barriers, regulatory uncertainties, and stakeholder diversity. While technical XAI research has advanced significantly, its translation into banking practice remains limited by socio-technical implementation gaps.

The primary contribution of this review is identifying and analyzing these implementation barriers while proposing integrative solutions. We move beyond cataloging XAI techniques to examining why promising technical solutions fail to achieve widespread adoption and how these failures might be addressed through combined technical, organizational, and regulatory approaches.

For practitioners, this review provides a framework for developing comprehensive XAI strategies that address both technical and human factors. For researchers, it identifies critical gaps in current understanding, particularly regarding longitudinal implementation processes, economic evaluation, and cross-stakeholder explanation design. For regulators, it highlights the need for clearer standards while maintaining flexibility for innovation.

Author contributions

CRediT: **Ntombikayise Betreace Tuge:** Conceptualization, Data curation, Investigation, Methodology, Writing – original draft; **Nkosikhona Theoren Msweli:** Formal analysis, Supervision.

Disclosure Statement

No potential conflict of interest was reported by the author(s).

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Data Availability Statement

The data supporting the findings of this study consist entirely of publicly available literature and secondary sources cited within the manuscript. No primary or proprietary datasets were generated or analyzed for this systematic review. All sources are detailed in the references section.

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